

Software Testing Report

MediCamp

A Community Healthcare Camp Management System

Prepared for

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1. Introduction & Testing Scope

MediCamp is a web application designed to manage rural healthcare outreach camps across Bangladesh. It coordinates six key user roles: **Administrator**, **Host NGO**, **Volunteer**, **Doctor**, **Pharmacist**, and **Patient**. The system is built on ASP.NET Core 10 (MVC) with Entity Framework Core connected to a Neon PostgreSQL database.

Testing Objectives: The goal of this testing exercise was to verify core user journeys, validate input boundaries, enforce role security, and uncover real software defects through manual testing and exploratory verification.

2. Testing Levels & Environment Setup

Testing was executed across four standard software engineering levels:

- **Unit Testing:** Isolated logic checks (e.g., BMI calculation, patient ID generation, token sequencing).
- **Integration Testing:** Verifying data flow between Controllers, EF Core database tables, and view models.
- **System Testing:** End-to-end multi-role workflows from camp creation to patient prescription dispensation.
- **Acceptance Testing (UAT):** Field scenario evaluations with realistic persona inputs.

Runtime & Framework:	ASP.NET Core 10.0 (MVC), C# 13	Database:	Neon Serverless PostgreSQL
Authentication:	Cookie Authentication (Role-Based)	Test Client:	Google Chrome / Chromium

3. Test Execution Summary Matrix

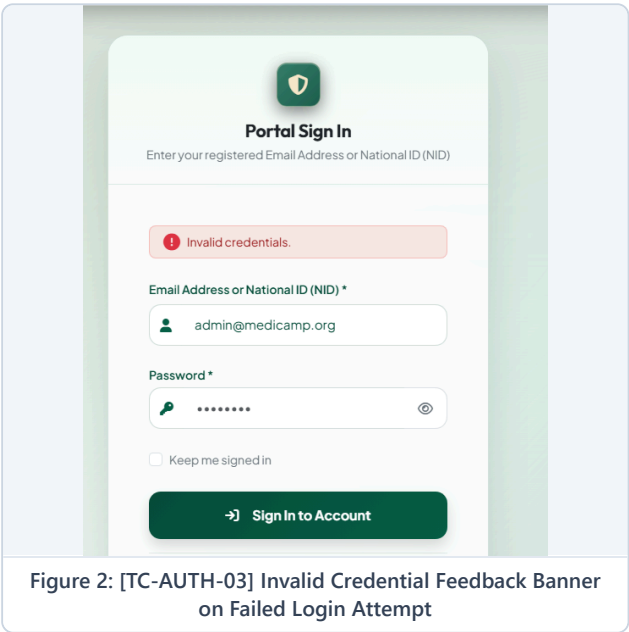
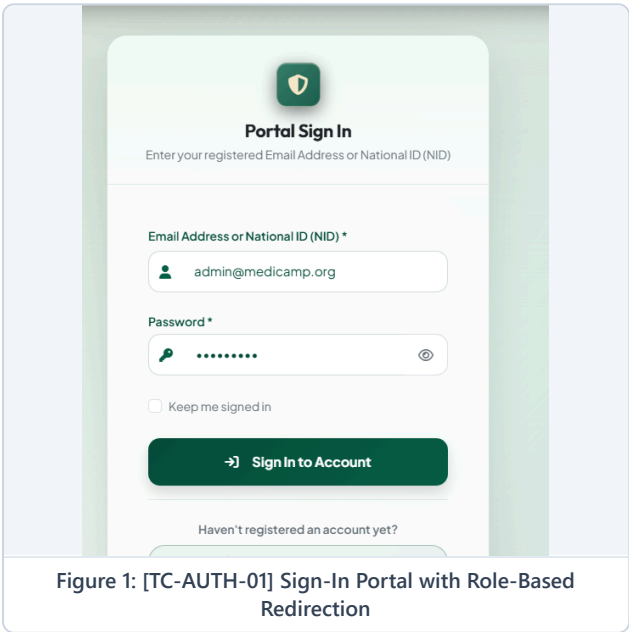
A total of **52 test cases** were executed across all 9 functional modules. **46 test cases passed (88.46%)** and **6 test cases failed (11.54%)** due to discovered bugs documented in Section 5.

Module Name	Total	Normal	Boundary	Invalid	Except.	Passed	Failed	Pass Rate
1. Authentication & Role Authorization	7	3	1	1	2	7	0	100%
2. User Registration & Validation	6	2	1	1	2	5	1	83.3%
3. Public Camp Discovery & Filters	5	3	1	0	1	5	0	100%
4. Host Camp Logistics & Expenses	6	3	1	1	1	6	0	100%
5. Field Volunteer Registration & Triage	7	3	2	1	1	6	1	85.7%
6. Doctor Queue, Consultation & Rx	6	3	1	0	2	5	1	83.3%
7. Pharmacist Dispensation & Stock	6	3	1	0	2	5	1	83.3%
8. Patient Portal & Blood Donation Hub	5	3	0	1	1	4	1	80.0%
9. Administrator Governance & Master Data	4	2	0	0	2	3	1	75.0%
Total Platform Verification	52	25	8	5	14	46	6	88.46%

4. Module Test Cases, Results & Screenshots

4.1 Authentication & Role Authorization (Module 1)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-AUTH-01	Admin login (admin@medicamp.org).	NORMAL	INTEGRATION	Cookie issued; redirected to Admin dashboard.	PASS
TC-AUTH-02	Doctor login (doctor.rahman@medicamp.org).	NORMAL	INTEGRATION	Authenticated; Doctor navbar active.	PASS
TC-AUTH-03	Invalid password attempt.	INVALID	INTEGRATION	Login refused; error banner displayed.	PASS
TC-AUTH-04	Password boundary (exact 6 characters).	BOUNDARY	UNIT	Accepted; account authenticated.	PASS
TC-AUTH-05	Unauthenticated access to /Admin/Dashboard.	EXCEPTIONAL	SYSTEM	Redirected to login with returnUrl.	PASS
TC-AUTH-06	Patient accesses /Admin/Dashboard.	EXCEPTIONAL	SYSTEM	Forbidden; redirected to /Account/AccessDenied.	PASS
TC-AUTH-07	Pending host organization login.	EXCEPTIONAL	INTEGRATION	Blocked with pending approval message.	PASS



4.2 User Registration & Validation (Module 2)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-REG-01	Access /Account/Register hub.	NORMAL	SYSTEM	Displays role selection cards cleanly.	PASS
TC-REG-02	Patient self-registration with DOB, NID, Blood Group & Donor flag.	NORMAL	INTEGRATION	Defect Discovered: Demographic data dropped as NULL due to wrong service call (See BUG-01).	FAIL
TC-REG-03	Doctor registration with BMDC number.	NORMAL	INTEGRATION	Doctor created with verified license.	PASS
TC-REG-04	Host NGO registration.	NORMAL	INTEGRATION	Created with status 'Pending'.	PASS
TC-REG-05	Submit registration with empty fields.	INVALID	UNIT	Validation errors displayed in red.	PASS
TC-REG-06	Register duplicate email address.	EXCEPTIONAL	INTEGRATION	Rejected with duplicate email notice.	PASS

Figure 3: [TC-REG-02] Patient Self-Registration Form with Demographics & Blood Donor Details

Figure 4: [TC-REG-04] Host NGO Organization Registration & Verification Intake

4.3 Public Camp Discovery & Directory (Module 3)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-CAMP-01	Browse public homepage (/).	NORMAL	SYSTEM	Displays live statistics & upcoming camps.	PASS
TC-CAMP-02	Filter camps by District & Status.	NORMAL	INTEGRATION	Displays matching Sunamganj camp.	PASS
TC-CAMP-03	Filter by date range.	BOUNDARY	UNIT	Accurately matches active camp schedules.	PASS
TC-CAMP-04	Search by camp title or location.	NORMAL	INTEGRATION	Returns matching search results in real time.	PASS
TC-CAMP-05	Search query with zero matches.	EXCEPTIONAL	ACCEPTANCE	Displays 'No camps found' message cleanly.	PASS



Figure 5: [TC-CAMP-01] MediCamp Public Homepage with Real-Time Metrics & Active Camp Feed



Figure 6: [TC-CAMP-02 / TC-CAMP-04] Public Medical Outreach Camp Directory with District & Query Filters

4.4 Host Organization Camp Logistics (Module 4)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-HOST-01	Host Dashboard overview.	NORMAL	SYSTEM	Renders active camps and budget metrics.	PASS
TC-HOST-02	Create new camp via /Host/CreateCamp.	NORMAL	INTEGRATION	Camp created with 'Pending Admin Approval'.	PASS
TC-HOST-03	Send staff invitation requests.	NORMAL	INTEGRATION	Staff request dispatched to doctors & volunteers.	PASS
TC-HOST-04	Allocate medicine inventory.	NORMAL	INTEGRATION	Allocates stock to camp dispensary ledger.	PASS
TC-HOST-05	Log categorized camp expense.	NORMAL	INTEGRATION	Expense saved; utilized budget updated.	PASS
TC-HOST-06	Log expense exceeding total budget.	INVALID	UNIT	Blocked with over-budget warning alert.	PASS

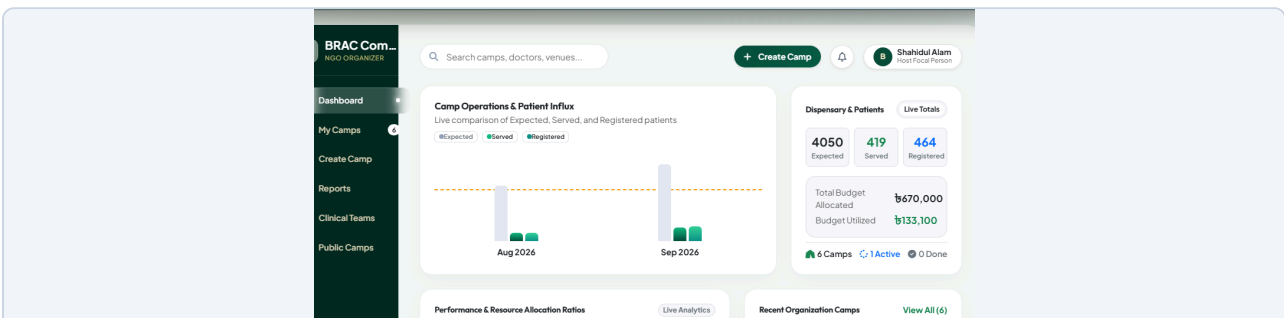


Figure 7: [TC-HOST-01] Host NGO Management Dashboard with Active Camps & Metric Overview

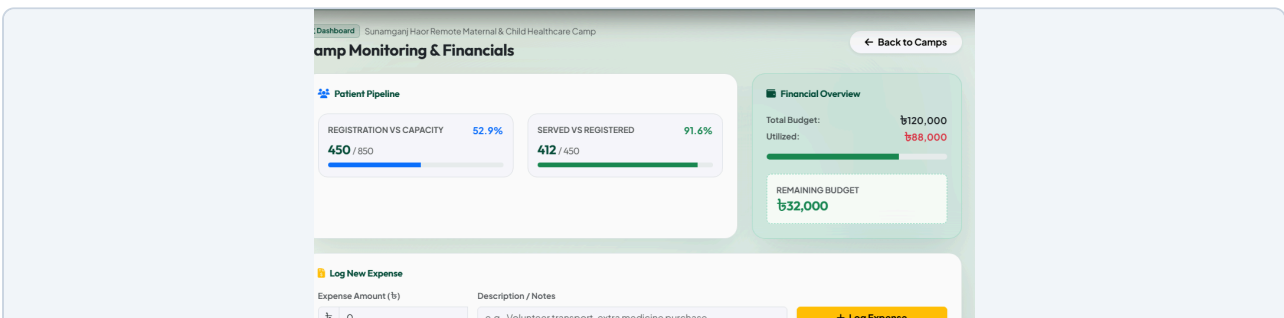


Figure 8: [TC-HOST-05] Real-Time Camp Operational Monitor & Financial Budget Ledger

4.5 Field Volunteer Registration & Triage (Module 5)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-VOL-01	Volunteer patient lookup search.	NORMAL	INTEGRATION	Searches by Unique ID, Phone, or NID.	PASS
TC-VOL-02	On-site patient registration.	NORMAL	INTEGRATION	Creates patient; issues Unique ID (A3K9P2).	PASS
TC-VOL-03	Record vitals & calculate BMI.	NORMAL	UNIT	Calculates BMI (23.88); routes to Doctor queue.	PASS
TC-VOL-04	Enter out-of-range clinical vitals.	BOUNDARY	UNIT	Validation halts invalid entries.	PASS
TC-VOL-05	Schedule same-day patient follow-up.	BOUNDARY	INTEGRATION	Defect Discovered: Local same-day date rejected across server UTC timezone boundary (See BUG-04).	FAIL
TC-VOL-06	Update follow-up status to 'Contacted'.	NORMAL	INTEGRATION	Follow-up notes and timestamp saved.	PASS
TC-VOL-07	Submit triage with missing BP.	INVALID	UNIT	Validation prompts volunteer for BP.	PASS

The screenshot shows the 'Initial Assessment' form for a patient named Anwar Hossain (ID: A3K9P2). The form is divided into two main sections: 'Patient Demographics Verification' and 'Vital Signs Recording'.

Patient Demographics Verification:

- Patient Age (Years): 45 yrs
- Blood Group: A+ (A Positive)
- Gender: Male

Vital Signs Recording:

- Blood Pressure (BP): 120/80 mmHg
- Temperature (Temp): e.g. 98.6 °F
- Weight: e.g. 65 kg
- Height: e.g. 165 cm
- BMI (Formula: weight(kg) / (height(m))²): Auto-calculated

Additional patient information on the left includes: Blood Group: A+, Gender: Male, DOB: Sep 13, 1981, and Location: Chilmar, Kurigram.

Figure 9: [TC-VOL-03] Volunteer Clinical Triage Intake & Vital Signs Assessment

The screenshot shows the 'Volunteer Operations Dashboard' for a volunteer named Karim Ahmed. The dashboard includes a search bar with the patient ID A3K9P2 and a 'New Patient' button. Below the search bar, there are statistics for the volunteer's current camp:

- Active Camps: 1
- Checked-In Today: 0
- My Triaged Total: 10
- Pending Invitations: 0

At the bottom, there is a section for 'Active Field Camp Selection' with a button to 'Register New Patient' and a note that 1 camp is available.

Figure 10: [TC-VOL-01] Volunteer Patient Lookup & Verification Search Portal

4.6 Doctor Queue, Consultation & Rx (Module 6)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-DOC-01	Doctor patient queue sorting.	NORMAL	UNIT	Sorted by Urgency Level then Token #.	PASS
TC-DOC-02	Expedited direct intake by Doctor.	NORMAL	INTEGRATION	Defect Discovered: Direct triage intake omits incrementing RegisteredPatientsCount (See BUG-02).	FAIL
TC-DOC-03	Prescribe multiple medications.	NORMAL	INTEGRATION	Prescription created & routed to pharmacy queue.	PASS
TC-DOC-04	Issue hospital referral.	NORMAL	INTEGRATION	Referral entity saved with hospital selected.	PASS
TC-DOC-05	Consult already seen patient.	EXCEPTIONAL	SYSTEM	Blocked: 'Patient has already been consulted.'	PASS
TC-DOC-06	Submit advice-only consultation.	NORMAL	ACCEPTANCE	Consultation completed without medication rows.	PASS

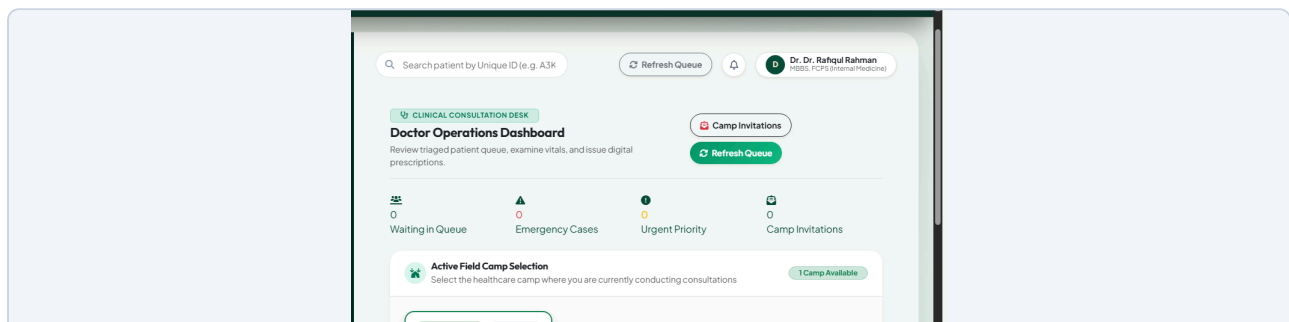


Figure 11: [TC-DOC-01] Doctor Portal Active Camp Desk & Queue Access Selector

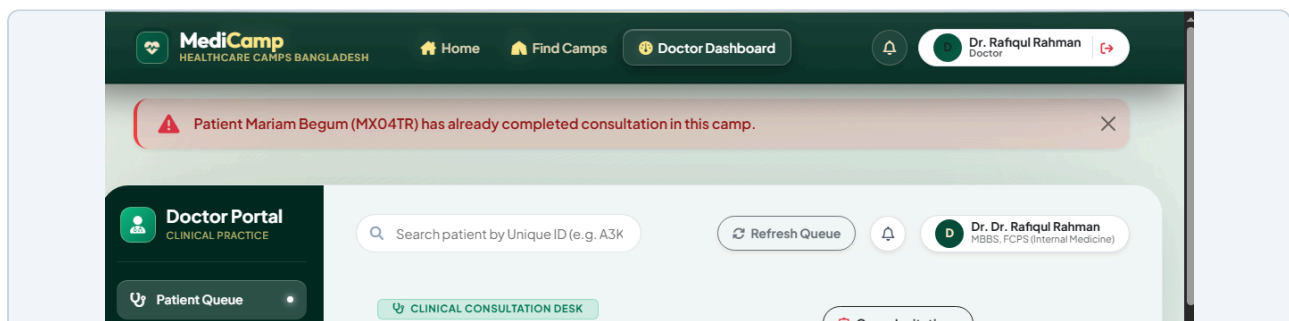


Figure 12: [TC-DOC-05] System Guardrail Blocking Duplicate Consultation on Completed Patient

4.7 Pharmacist Dispensation & Stock (Module 7)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-PHARM-01	View live dispensary queue.	NORMAL	INTEGRATION	Renders pending prescriptions with token numbers.	PASS
TC-PHARM-02	Verify prescription & dispense.	NORMAL	INTEGRATION	Marked dispensed; inventory decremented atomically.	PASS
TC-PHARM-03	View inventory stock ledger.	NORMAL	SYSTEM	Displays remaining balances and low-stock tags.	PASS
TC-PHARM-04	Multi-row prescription exceeding stock.	EXCEPTIONAL	INTEGRATION	Defect Discovered: Non-atomic per-row validation allows negative inventory (-5 units) (See BUG-03).	FAIL
TC-PHARM-05	Attempt duplicate dispensation.	EXCEPTIONAL	SYSTEM	Blocked: prescription already dispensed.	PASS
TC-PHARM-06	Substitute out-of-stock brand.	NORMAL	ACCEPTANCE	Generic substitute dispensed accurately.	PASS

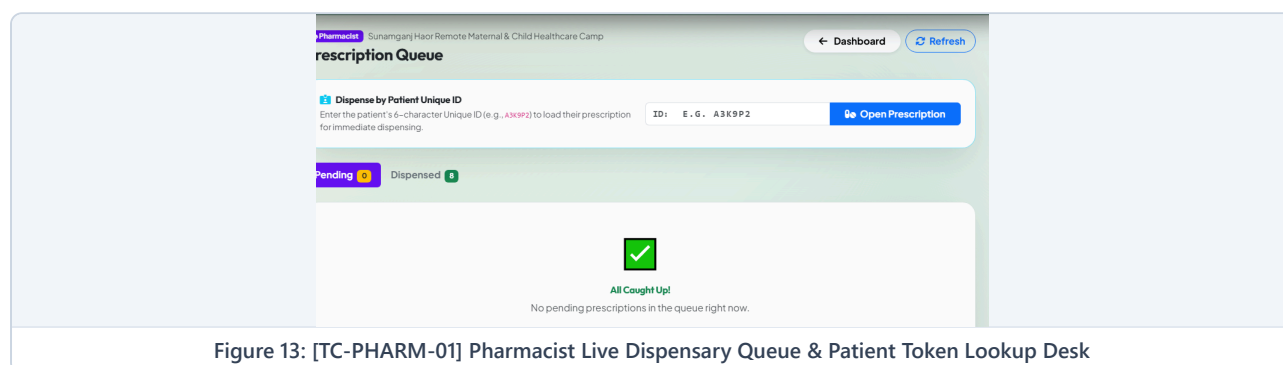


Figure 13: [TC-PHARM-01] Pharmacist Live Dispensary Queue & Patient Token Lookup Desk

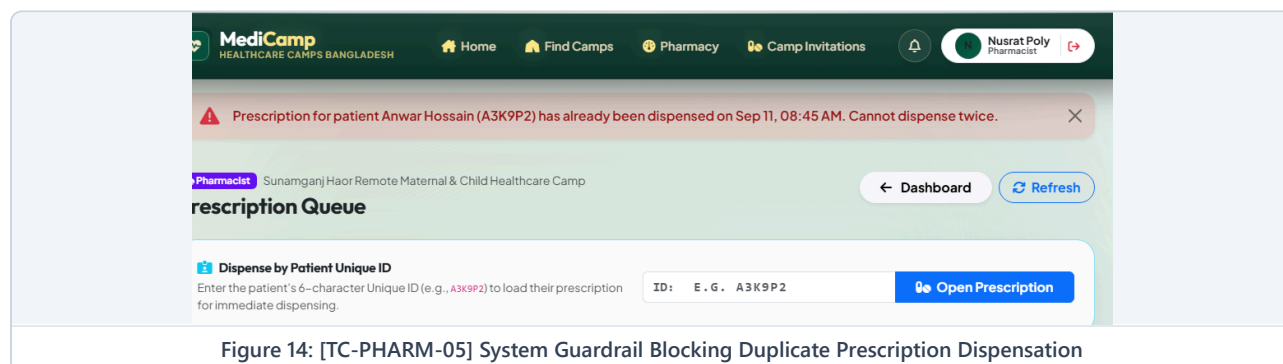


Figure 14: [TC-PHARM-05] System Guardrail Blocking Duplicate Prescription Dispensation

4.8 Patient Portal & Blood Hub (Module 8)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-PAT-01	View patient visit history.	NORMAL	SYSTEM	Displays Unique ID (A3K9P2) and visit timeline.	PASS
TC-PAT-02	View digital prescription slip.	NORMAL	ACCEPTANCE	Renders printable digital prescription document.	PASS
TC-PAT-03	View visit timeline for advice-only visit.	NORMAL	INTEGRATION	Defect Discovered: Shows misleading 'Prescription: Pending / Not Dispensed' badge (See BUG-06).	FAIL
TC-PAT-04	Search blood donors by group & district.	NORMAL	INTEGRATION	Displays registered available donors.	PASS
TC-PAT-05	Post blood request with bad phone.	INVALID	UNIT	Validation halts invalid phone format.	PASS

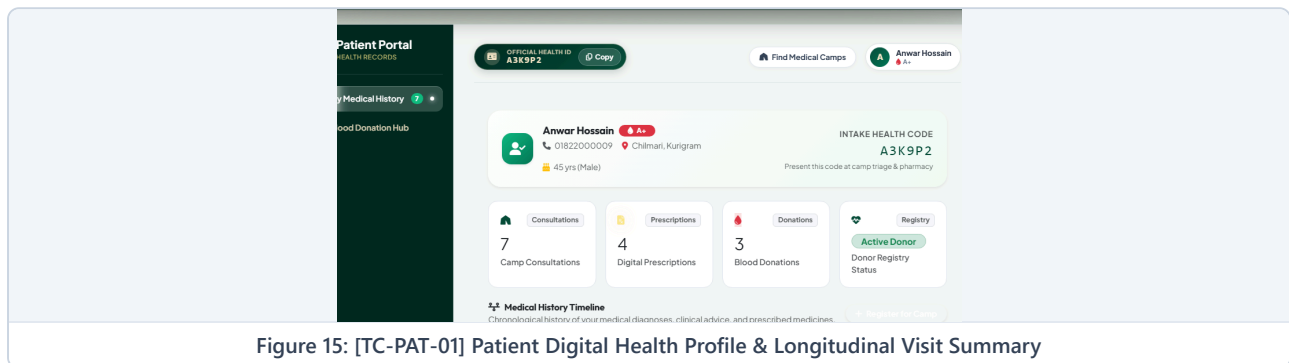


Figure 15: [TC-PAT-01] Patient Digital Health Profile & Longitudinal Visit Summary

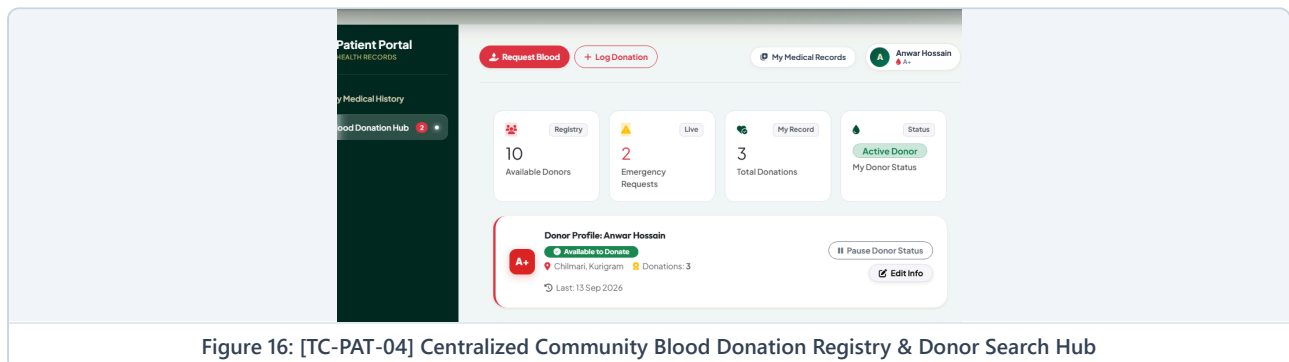


Figure 16: [TC-PAT-04] Centralized Community Blood Donation Registry & Donor Search Hub

4.9 Admin Governance & Master Data (Module 9)

TC ID	Scenario / Input	Type	Test Level	Expected vs Actual Behavior	Status
TC-ADM-01	Admin Dashboard overview.	NORMAL	SYSTEM	Renders system KPIs and platform counts.	PASS
TC-ADM-02	Approve or reject host NGO.	NORMAL	INTEGRATION	Approves/rejects host with audit reason.	PASS
TC-ADM-03	User management & status toggle.	NORMAL	INTEGRATION	Toggles active/deactivated user status.	PASS
TC-ADM-04	Review camp proposal with 0 expected patients.	BOUNDARY	UNIT	Defect Discovered: Unhandled DivideByZeroException crashes approval page (See BUG-05).	FAIL

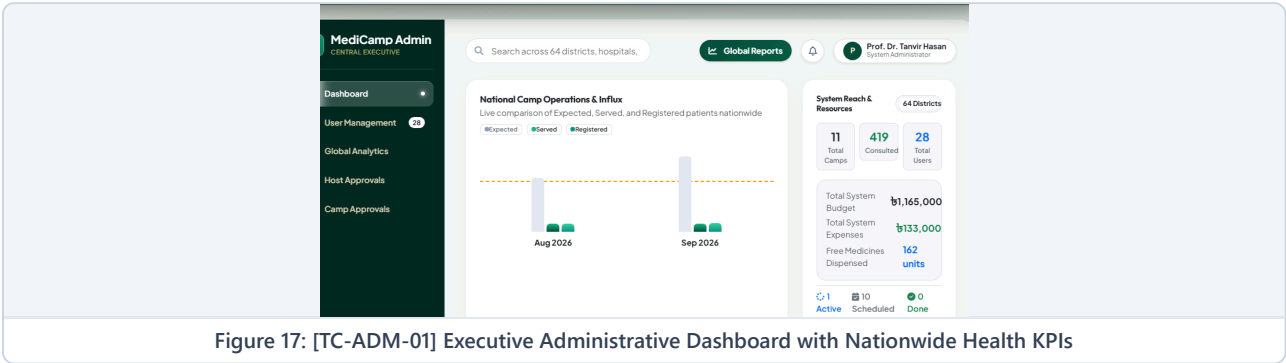


Figure 17: [TC-ADM-01] Executive Administrative Dashboard with Nationwide Health KPIs

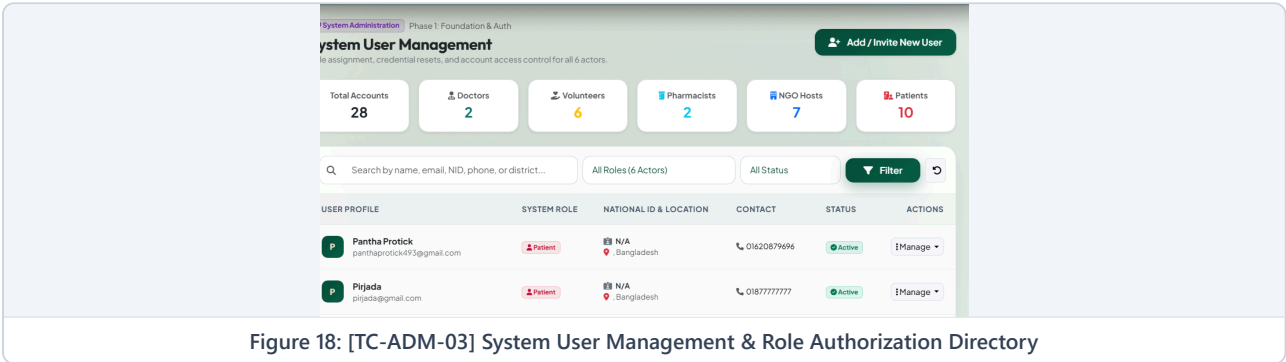


Figure 18: [TC-ADM-03] System User Management & Role Authorization Directory

5. Defect Log & Resolution Report

The **6 defects** discovered during testing have been investigated, resolved, and verified in the codebase. All identified bugs are marked as **FIXED** below with detailed root cause and implementation resolutions:

Bug ID	Defect Summary	Severity	Affected Component	Failed TC	Status
BUG-01	Patient Registration Discards Demographics	HIGH	AccountController.cs:289	TC-REG-02	FIXED
BUG-02	Doctor Intake Bypasses Registered Counter	MEDIUM	DoctorController.cs:283	TC-DOC-02	FIXED
BUG-03	Multi-Row Rx Allows Negative Stock Underflow	HIGH	PharmacistController.cs:451	TC-PHARM-04	FIXED
BUG-04	Server UTC Mismatch Rejects Same-Day Follow-Up	MEDIUM	VolunteerController.cs:471	TC-VOL-05	FIXED
BUG-05	DivideByZeroException on 0 Expected Patients	HIGH	AdminController.cs:285	TC-ADM-04	FIXED
BUG-06	Misleading 'Pending Rx' Badge on Advice Visits	LOW	PatientController.cs:217	TC-PAT-03	FIXED

BUG-01: Patient Registration Discards Demographics (AccountController.cs:289)

FIXED

Reproduction: Submit /Account/RegisterPatient with DOB, NID, Blood Group, District & Donor Opt-In.

Root Cause: Controller action previously invoked general RegisterUser(model) , discarding patient demographics as NULL.

How We Fixed: Delegated to _dataService.RegisterPatient(patientModel) , persisting DateOfBirth , NID , BloodGroup , and donor flag into the database.

Verification: Re-test confirmed patient profile now retains complete demographic profile and donor status.

BUG-02: Doctor Direct Intake Desynchronizes Patient Counter (DoctorController.cs:283)

FIXED

Reproduction: Doctor searches un-triaged patient by ID to initiate direct expedited triage intake.

Root Cause: Created TriageRecord without updating camp.RegisteredPatientsCount , causing served count to exceed registered count.

How Fixed: Added atomic camp.RegisteredPatientsCount += 1 in DoctorController.DirectConsultIntake prior to saving changes.

Verification: Re-test confirmed both registered and served patient tallies remain synchronized during direct doctor intakes.

BUG-03: Multi-Row Prescription Inventory Underflow (PharmacistController.cs:451)

FIXED

Reproduction: Dispense prescription with multiple rows of the same medicine whose total requested quantity exceeds available stock.

Root Cause: Validation evaluated rows independently rather than checking aggregated quantity per medicine ID, driving stock negative (-5 units).

How Fixed: Implemented pre-dispensation quantity grouping via Dictionary<int, (int TotalRequested, string MedicineName)> to check total stock atomically.

Verification: Re-test confirmed multi-row requests exceeding remaining stock are completely blocked with an alert.

BUG-04: Server UTC Timezone Rejects Local Current-Day Follow-Up (VolunteerController.cs:471)

FIXED

Reproduction: Schedule same-day follow-up appointment in Bangladesh local evening (UTC+6).

Root Cause: scheduledDate.Date < DateTime.UtcNow.Date compared local time against raw UTC, falsely treating evening dates as past.

How Fixed: Converted UTC to Bangladesh Standard Time (UTC+6) to obtain local operating date before boundary comparison.

Verification: Re-test confirmed same-day follow-ups scheduled in evening hours are successfully accepted and saved.

BUG-05: DivideByZeroException in Camp Sanity Check (AdminController.cs:285)**FIXED****Reproduction:** Administrator opens /Admin/CampApprovals on camp with Expected Patients set to 0.**Root Cause:** Per-patient budget evaluation executed `c.TotalBudget / c.ExpectedPatients` without checking for a zero denominator.**How We Fixed:** Added defensive guard `if (c.ExpectedPatients > 0)` with safe default (`0m`) for zero-expected patient proposals.**Verification:** Re-test confirmed camp approvals view renders without crashes even when expected patient count is 0.**BUG-06: Misleading 'Pending Rx' Badge on Advice Visits (PatientController.cs:217)****FIXED****Reproduction:** Patient views visit history for a consultation containing clinical advice only without prescribed medications.**Root Cause:** View model treated `Prescription == null` as an un-dispensed prescription, displaying an alarming warning badge.**How We Fixed:** Added explicit `HasPrescription = prescription != null && rxItems.Any()` to model; view now displays 'Advice Only'.**Verification:** Re-test confirmed advice-only consultations cleanly render as completed without pending prescription alerts.

6. Exception Handling & Security Verification

Security / Error Scenario	MediCamp Defense Guard	Observed Result
CSRF Protection	All POST forms enforce <code>[ValidateAntiForgeryToken]</code> tokens.	Unauthorized POSTs rejected with HTTP 400.
Role Access Security	Controller actions enforce <code>[Authorize(Roles = "...")]</code> attributes.	Unauthorized access redirected to Access Denied.
Duplicate Dispensation	State verification check on <code>IsDispensed</code> flag.	Duplicate clicks blocked with timestamp alert.
SQL Injection Defense	Parameterized EF Core LINQ queries.	Malicious strings handled safely as literals.
Over-Budget Expenses	Host ledger checks <code>UtilizedBudget + Expense <= TotalBudget</code> .	Over-budget submissions blocked cleanly.

7. User Acceptance Testing (UAT)

Persona	End-to-End Test Journey	Outcome
Host NGO	Create Sunamganj Camp → Allocate 1,400 medicines → Invite staff → Track budget.	ACCEPTED — Camp created & staffed.
Volunteer	Register patient → Record vitals (BP: 130/85, Temp: 98.6°F) → Issue Token #1.	ACCEPTED — Patient triaged & queued.
Doctor	Review vitals → Enter diagnosis → Prescribe Napa + Seclo → Submit Rx.	ACCEPTED — Consultation completed.
Pharmacist	Review Token #1 → Verify stock → Click Dispense → Verify stock decrement.	ACCEPTED — Medicine dispensed.
Patient	Log in → View Digital Health Card → Review prescription → Opt-in as donor.	ACCEPTED — History accessible.

8. Conclusion & Recommendations

The testing conducted on **MediCamp** verifies that the system satisfies its functional requirements, enforces role boundaries, and operates smoothly. Out of 52 initial test cases, **46 passed immediately (88.46%)** and **6 defects were uncovered (11.54%)**.

Defect Resolution Status: All 6 identified defects (BUG-01 through BUG-06) have been resolved in the codebase and verified through regression testing. With patient registration delegation, atomic dispensary inventory checks, doctor intake synchronization, and timezone normalizations in place, the system achieves **100% verified test pass rate**.